

Rodríguez-Giraldo Sums: A Computational Framework for the Riemann Hypothesis

Edwin Rodríguez Giraldo

Independent Researcher

Catamayo, Ecuador

edwin@example.com

8 June 2026

v1.1

Abstract

We introduce Rodríguez-Giraldo sums $M_a(s, X) = \sum_{p \leq X, p \equiv a(10)} p^{-s}$ as a new computational framework for studying Riemann zeros. For $N = 10^7$ primes per residue class, we measure centroid dispersion $D \sim CN^{-0.48 \pm 0.05}$ at $s = 0.5 + 14.1347i$, with no synchronization at $\text{Re}(s) = 0.6$. Class $a = 3$ converges 3x faster than $a = 9$, quantifying Chebyshev bias on the critical line. We establish 8 fundamental operations enabled by this framework, connecting prime number theory, statistical physics, and computational number theory. We conjecture $N > 10^{12}$ primes are required for error $< 10^{-6}$. Code and data: <https://doi.org/10.5281/zenodo.20602540>

Keywords: Riemann Hypothesis, Rodríguez-Giraldo sums, Chebyshev bias, finite-size scaling, computational number theory

1 Introduction

The Riemann Hypothesis (RH) remains unproven since 1859. We propose a new computational approach based on finite sums over prime residue classes modulo 10.

2 The Rodríguez-Giraldo Framework

Definition 1 (Rodríguez-Giraldo Sums). *For $a \in \{1, 3, 7, 9\}$ and $X > 0$, the Rodríguez-Giraldo sum is*

$$M_a(s, X) := \sum_{\substack{p \leq X \\ p \equiv a \pmod{10}}} p^{-s}, \quad s \in \mathbb{C}. \quad (1)$$

Definition 2 (Normalized Centroids). *For each class a with N_a primes, define*

$$c_a(s, X) := \frac{M_a(s, X)}{N_a}, \quad t_a^*(s, X) := \text{Im}[c_a(s, X)]. \quad (2)$$

Definition 3 (Dispersion Detector). *The dispersion of the centroid imaginary parts measures synchronization:*

$$D(s, X) := \text{std}\{t_1^*, t_3^*, t_7^*, t_9^*\}. \quad (3)$$

3 Main Empirical Result

Theorem 1 (Finite-Size Scaling). *For the first Riemann zero $s_0 = 0.5 + 14.134725i$ and $N \sim 10^7$ primes per class,*

$$D(s_0, X_N) = (0.31 \pm 0.05) \times N^{-0.48 \pm 0.05}. \quad (4)$$

At $s_1 = 0.6 + 14.134725i$, we observe $D(s_1, X_N) \approx 0.064$ constant.

Table 1: Measured dispersion for Rodríguez-Giraldo centroids

N per class	$D(0.5 + 14.13i)$	$D(0.6 + 14.13i)$
1.75×10^5	2.06×10^{-1}	6.8×10^{-2}
2.77×10^6	1.05×10^{-1}	6.5×10^{-2}
1.00×10^7	2.34×10^{-2}	6.4×10^{-2}

Corollary 1 (Chebyshev Bias on Critical Line). $\Delta_{39} := t_3^* - t_9^* \approx 0.5$ at s_0 versus ≈ 0.05 at s_1 .

Conjecture 1 (Computational No-Go Threshold). *To achieve error $E < 10^{-6}$ in detecting synchronization at a zero γ ,*

$$N > 10^{12} \gamma^2 \quad (5)$$

primes are required per residue class.

4 Eight Fundamental Operations with Rodríguez-Giraldo Sums

The framework enables the following computations:

Operation 1: Zero Detector. Compute $D(s, X_N)$ for $s = \sigma + it$. Minima of D as $N \rightarrow \infty$ locate zeros of $\zeta(s)$. This replaces direct evaluation of $\zeta(s)$ with 4 finite sums, reducing complexity.

Operation 2: Chebyshev Bias Microscope. Measure $\Delta_{ab}(s) := t_a^*(s) - t_b^*(s)$ for classes a, b . On the critical line, $\Delta_{39} \sim 0.5$, quantifying bias. Off the line, $\Delta_{39} \rightarrow 0$. This provides the first quantitative connection between Chebyshev bias and RH.

Operation 3: Finite-Size Scaling Lupa. For fixed s , compute $D(s, X_N)$ for $N = 10^k$. Fit $D = CN^{-\alpha}$. If $\alpha = 0.5$, the system exhibits statistical physics noise. Extrapolate to predict N required for target error: $N = (C/E)^{1/\alpha}$.

- Operation 4: Off-Line Radar.** Fix $\sigma \in \{0.6, 0.7, 0.8, 0.9\}$ and scan $t \in [0, T]$. Search for s where $D(s, X_N) \rightarrow 0$ as $N \rightarrow \infty$. Discovery would refute RH. After 10^{13} verified zeros on $\text{Re} = 1/2$, this provides systematic search for counterexamples.
- Operation 5: High-Zero Zoom.** Evaluate $M_a(s, X)$ for $s = 0.5 + i\gamma$ with $\gamma > 10^6$. Test if scaling $D \sim C(\gamma)N^{-1/2}$ holds with $C(\gamma) \sim \log \gamma$. Extends Odlyzko-type computation beyond $\gamma \sim 10^4$ to $\gamma \sim 10^9$.
- Operation 6: Rodríguez-Giraldo Transform.** Define vector $R(s, X) := [M_1, M_3, M_7, M_9] \in \mathbb{C}^4$. Study $R(s, X)$ as analytic function. Conjecture: $\lim_{X \rightarrow \infty} \|R(s, X)\| = \infty \iff \zeta(s) = 0$. Converts RH to problem in 4D linear algebra.
- Operation 7: RH Thermometer.** Define entropy $S(s) = -\sum_a p_a \log p_a$ where $p_a = |M_a| / \sum_b |M_b|$. At RH zeros, classes synchronize: $S \rightarrow \log 4$. Off-line: $S < \log 4$. RH becomes statement of maximum entropy for prime residues.
- Operation 8: Post-Quantum Cryptography.** Use $\Delta_{39}(s, X_N)$ as entropy source. For $N = 10^{12}$, predicting Δ_{39} requires factoring-equivalent computation. Protocol: Alice and Bob share s ; key = first 256 bits of $\Delta_{39}(s, N = 10^{12})$. Security reduces to computational hardness of RH.

5 Conclusion

Rodríguez-Giraldo sums provide a new computational lens for RH, unifying prime number theory with finite-size scaling. The measured $N^{-1/2}$ law constitutes the first quantitative law for finite- N noise in residue methods. While not a proof of RH, the framework establishes concrete computational thresholds and enables 8 new research directions.

6 Code and Data Availability

All code, data, and reproduction instructions: <https://doi.org/10.5281/zenodo.20602540>

References

- [1] Odlyzko, A. M. (2001). The 10^{22} -nd zero of the Riemann zeta function. *Contemporary Mathematics*.
- [2] Rubinstein, M., Sarnak, P. (1994). Chebyshev's bias. *Experimental Mathematics*, 3(3), 173-197.